Luis Victoria

New York City | luisvictoriaperez@gmail.com | +1 (929) 420-8800 | linkedin.com/in/luisvictoria | github.com/lv

Work Experience

Software Engineer — Market Data Platform, Capacity & Performance Feb 2023 - Jul 2026
Bloomberg LP *New York, NY*

Engineered data, performance, capacity, scalability, and reliability systems across Bloomberg's Market Data Platform, spanning Feeds and the Ticker Plant. The platform handled >700B ticks/day during extreme volatility while maintaining <6ms median and <250ms p99.9 latency SLAs.

- **Performance Regression Detection**: Applied Bayesian Online Changepoint Detection within Argo-orchestrated workflows to statistically flag anomalous times across 30,000+ tick processor instances, cutting remediation time from days to <24 hours.
- **Latency Instrumentation Overhaul**: Authored a C++20 envelope-based timestamping framework tracing tick updates stage-by-stage through the Ticker Plant pipeline with μs resolution, enabling latency regression detection and optimization with $< 10\mu$s overhead.
- **Predictive Capacity Forecasting**: Rewrote a fragile legacy log-scraping storage monitor as a two-part Python system: a fleet-wide scanner writing structured per-database storage and machine metadata to a centralized datastore, and a forecasting service extrapolating rolling usage trends 14 days ahead with automated alerting. Delivered in one week during the April 2025 tariff-driven volatility spike, protecting 1,000+ machines from storage-exhaustion/data-loss risk.
- **Production & Incident Ownership**: Participated in 24/7 follow-the-sun support for Bloomberg's market-data platform, isolating bad data, performing live synchronization and recovery, and helping drive major outage response under volatile market conditions.
- **Capacity & Load Management**: Assessed CPU, memory, and storage capacity ahead of onboarding new global exchanges and executed targeted cluster splits to offload overloaded machines, avoiding or eliminating SLO breaches with minimal client disruption.
- **Disaster Recovery Performance Resilience**: Automated load reassignment during data-center failover; benchmarked CPU thread-pinning under combined load, informing Linux cgroups-based CPU prioritization to preserve broadcast continuity while controlling dropped requests.

Quantitative Developer Intern May 2019 - Aug 2019
Farringdon Asset Management *Singapore*

- **Automation with Bloomberg API**: Reduced time to generate client portfolio analysis by 50% for the portfolio management team using information from PORT<GO>.
- **Back-testing Enhancements**: Extended the firm's Python backtesting engine to support cryptocurrency analysis, integrating market-data pipelines through CCXT and Binance.

Education

Business and Computer Science Sep 2018 - Dec 2022
Bachelor of Commerce - Sauder School of Business, University of British Columbia *Vancouver, BC*

- **Relevant Coursework**: Machine Learning & Data Mining, Matrix Algebra, Calculus III, Empirical Economics, Advanced Algorithmic Design, Advanced Relational Databases.
- **Sauder Trading Simulation**: Built a Raspberry Pi arbitrage bot for a primitive non-matching order-book simulation, polling every two minutes and executing crossed bid/ask opportunities; placed 1st of 136 students.